

Why Internet Measurement is Needed

The Internet is a critical part of our national infrastructure, essential to daily life, national security, the economy, healthcare, education, and most other aspects of modern society. In one example highlighting that ubiquitous Internet access is in our national interest, the Biden Administration recently invested 65 billion dollars in expanding broadband infrastructure. The Internet's role in society means that understanding, maintaining, and improving its performance, reliability, and security are of the utmost importance, as there remain acute limitations on all these axes. Further, as our uses and demands of the Internet evolve, the infrastructure and protocols that power it are continuously evolving to deliver them.

Given the Internet's important role, limitations, and constant evolution, measurements are fundamental to understanding and improving its function. Internet measurement provides data to understand when the Internet is working as expected and, more importantly, helps to identify problems, to determine what needs to be changed, and to guide the deployment of new technologies. Decision makers at federal and state agencies, advocates, Internet governance organizations, and more all depend upon data produced by Internet researchers to decide how to allocate their resources and set their institutional agendas. Far from being isolated, the field of Internet measurement is directly responsible for informing how the Internet evolves in a data-driven, fact-based way. The Internet is not just an engineered system but instead is shaped by technical, economic, political, and societal concerns, with complex, competing, and evolving incentives, regulations, and business practices. So, it is extremely important to have measurements funded by public programs in the interest of understanding and bettering the Internet for society.

However, the same pace of change and scale of traffic that makes the Internet essential to understand also makes it difficult to measure. Internet measurement experiments are often complex and resource-intensive, requiring ongoing support for infrastructure for measurement and data storage, training and collaboration, and iterative development. Additionally, because the Internet is always changing and is inherently shaped by geographic, economic, and governmental factors, it is not straightforward to ensure that any given experiment can be replicated over time and/or in different areas of the network. As the Internet evolves, the field is presented with new requirements for measurements and, hence, the need for new measurement techniques and studies. Each of these constraints is compounded by the need to manage a finite set of resources.

The discussions about Internet measurement at this workshop built upon discussions and presentations at the two-part 2021 NSF Workshop on Overcoming Measurement Barriers to Internet Research (WOMBIR 2021), the resulting workshop report, and the NSF's Internet Measurement Research: Methodologies, Tools, and Infrastructure (IMR) program that followed the WOMBIR workshop. The WOMBIR report covers important open challenges in Internet measurement research in more detail than we can include in this section of this report as well, and so we intend this report to supplement rather than supersede the WOMBIR report. Rather than aiming for completeness, we highlight a subset of themes we deem important in light of the existing context of the WOMBIR report and IMR program.

Internet Measurement Isn't Applied Networking Research

While the Internet is composed of interconnected networks, Internet measurement is fundamentally different from other networking research. At the technical level, the systems built for Internet measurement research must scale to take global measurements and operate across diverse, error-prone environments, safely and accurately accounting for discrepancies in standards, device configurations, and network behavior while having little to no ground truth data to compare against or control of in-network devices — rather than managing or measuring a small number of controlled networks. In many cases, Internet measurement systems are designed to safely measure environments through network side channels where direct measurement is not possible and, in some cases, have adversarial interference (e.g., understanding the behavior of the Great Firewall of China). As a result, a distinct class of tools is used for understanding the Internet as a whole as opposed to individual networks. In addition, experiments can span the course of months to years to understand how the Internet is evolving, requiring researchers to build infrastructure to longitudinally record and analyze Internet behavior rather than making decisions in close to real-time as is needed for management.

Many Internet measurement questions are also not purely technical and are tied to understanding broader social questions on issues such as net neutrality, broadband Internet access, and privacy. Often, Internet measurement research guides standards, policy, law, and funding, rather than the needs of individual operators (e.g., helping to understand the deployment of new protocols), and uncovers misbehavior of Internet companies (e.g., certificate authorities and ISPs), rather than focusing on fundamental networking questions around the performance and security of a single network. Analysis focuses not only on understanding the specific measurements taken, but what these tell us about the efficacy of new policies, standards, and laws.

The Internet is Hard to Measure—and It’s Becoming Harder!

The Internet is inherently hard to measure and understand. The core protocols that define its behavior and are the basis for (almost) every measurement that can be done on the Internet were not designed to be instrumented or measured. In fact, many of the protocol design choices date to an era when it was reasonable to get in touch with an operator from any site that was causing problems as a step towards remediation. In that setting, the Internet was relatively simple to understand, and measurements were less critical, and so the protocols were not designed with measurement support in mind. These protocol limitations mean that it is not just researchers who do not have a full picture, as the limited visibility forces network operators to spend lots of time trying to explain and fix problems.

The structure of the Internet further complicates its understandability and measurement. It is made up of tens of thousands of independently operated networks. Any use of the Internet will traverse multiple independent networks, and so no one has full access or visibility into any Internet transaction, much less into the full Internet. This complexity and distributed management means that measuring the Internet to understand its structure and behavior is a much bigger step than for networks or systems under unified control (such as a data center run by Meta).

The distributed management also complicates improvements to the protocols, as the networks must interoperate for the Internet to work, and so changes require coordination among networks with competing interests with millions of active network devices depending on the protocols in their current versions. This setting creates a situation where protocol improvements face nearly insurmountable barriers to adoption. The situation can be even more complicated for improvements to improve visibility or measurement support, as incentives for adopting such changes tend to be misaligned—changes made in one network help other networks understand that network but do not benefit the network itself.

On top of these inherent obstacles to measurement, the Internet is becoming simultaneously larger and more opaque. Numerous recent trends contribute to hiding the Internet from existing measurement techniques, including: Increasingly more Internet traffic is contributed by a small number of cloud and content providers, and they purposefully hide aspects of the private infrastructure they use to host services. The Internet has “flattened,” with the networks hosting users and servers interconnecting directly with each other rather than relying on a hierarchical architecture of intermediate regional, national, and intercontinental networks. These direct “peering” links are usually invisible to traditional measurements and available vantage points. A growing number of services encrypt their communication, obfuscating Internet traffic to passive measurement techniques. More and more web pages and web services require logs in and are customized to users and geolocations, hiding them from web-crawling measurements. As Internet of Things devices and mobile applications play a growing role on the Internet, a decreasing share of Internet behavior can be characterized by the behavior of a small number of platforms and web browsers that support general web-access behavior (and hence can be characterized by a small number of measurements towards sites of the researcher’s choice). Please refer to Section 3 of the WOMBIR report for further discussion of these and other trends challenging visibility.

Challenges in Supporting Measurements Across Time and Space

The value in much Internet measurement research is in the results, but the Internet can change quickly and unpredictably, meaning that published results go stale quickly—one cannot know whether they still reflect the state of the Internet—and measurements, like cars, lose much of their value as soon as they are driven off the lot. Those who rely on measurements—network and service operators, researchers, regulators, policy-makers, etc.—need up-to-date, public, longitudinal data, but instead the incentives for publication, funding, and graduation/promotion often favor one-off snapshots that may quickly be stale. Program committees favor novelty rather than revisiting previous results, especially if those previous results have not changed (which one cannot generally know without measuring). Most funding sources fund periods of at most a handful of years, which can be too short to understand Internet trends, and most calls favor innovation and funding at the granularity of a “grad student year,” rather than the smaller amounts of money over longer periods of time needed to maintain measurement projects after their initial development. The need for graduate students to publish and graduate gives an incentive to abandon projects after publication. However, if existing results are not publicly refreshed, one cannot know which remain “safe” to rely on. Efforts, including Community Awards, reproducibility badges, and poster Reproducibility Track, intend to encourage public data and revisiting of results, but most are assessed at submission time and provide limited incentive compared to the opportunity cost. Please refer to Section 4 of the WOMBIR report for further discussion of longitudinal measurements.

The Internet is also expanding: more users, in more parts of the world, using more services for more time with more traffic. Many studies require in situ or well-placed measurement vantage points, but achieving broad measurement and vantage point coverage—and maintaining that coverage over time—faces similar funding and incentive challenges as longitudinal measurements, as explained in the previous paragraph.

As the Internet becomes increasingly privatized, the Internet measurement data that is wanted and needed by researchers is often collected by private companies. However, there is no incentive structure in place that would make sharing that data in a private entity's best interest. Companies often consider their data proprietary by default and, therefore, consider publishing data openly an exception rather than the rule. When they do share data, it can be useful to researchers with an aligned use case, but is often aggregated or anonymized in a way that makes it difficult to replicate or use to produce more fine-grained or customized experiments that diverge from the company's intended use. Even when motivated to share more data in the open, protecting user privacy is a valid and shared interest – no research study would ultimately benefit from the exploitation of user data. However, no standard methodology or infrastructure exists to allow companies to share data with ease and confidence that it will protect their users. As the Internet becomes increasingly operated by private entities, removing barriers to collaboration and data sharing is essential. Without their cooperation and the development of an open data-sharing ecosystem that includes data that was initially collected privately, academic researchers will be working with a severely incomplete picture of the Internet. Please refer to Section 5 of the WOMBIR report for further discussion of privacy and Section 6 for further discussion of incentives.

Recommendations

Please refer to Section 7 of the WOMBIR report for further discussion of recommendations, with which the workshop attendees largely agreed. The recommendations in this section are intended to build on and highlight aspects of those recommendations, especially in light of the NSF's Internet Measurement Research: Methodologies, Tools, and Infrastructure (IMR) program that came after the WOMBIR report.

Many of the challenges described are the result of researchers using ad hoc methods to achieve their goals and complete short-term projects. Internet measurement is an iterative process, and valuable time and resources can be saved by building upon past efforts. We recommend that NSF provide incentives and resources for long-term, standardized, and open efforts to emerge, specifically: Development of open datasets created by and for scientific measurement research Development of privacy-conscious open data-sharing ecosystems to enable private data to be used for research in an ethical manner Maintenance and scaling of existing measurement infrastructure and tools Skill development for researchers to build more sustainable measurement systems

Additionally, we recommend that NSF prioritize efforts to standardize measurement methods. Interventions could include workshops to build consensus around the utility or definition of specific metrics, software development for consistent measurement and data collection, research to identify best practices, documentation of and reflection upon failed experiments that should not be included in standards, and more. Standardization is a resource-intensive, challenging, time-consuming process, but can increase the efficiency of the ecosystem and trust within the community.

Workshop attendees and the broader Internet measurement community view NSF's Internet Measurement Research: Methodologies, Tools, and Infrastructure (IMR) program as a great success, funding dozens of high-quality projects during its initial two-year period. Having this new NSF program, with a call and review panels focused on Internet measurement, invigorated the research area and demonstrated its importance and commitment to the area by the government, NSF, and the research community. This separate call for Internet measurement proposals (versus broader networking or security calls where some proposals might include or focus on Internet measurement) facilitated proposal review by subject matter experts, based on proposals' potential for successful real-world measurements and real-world impact, rather than tools, techniques, or systems being the sole end goal. The projects funded by the program have and are resulting in advancements, research papers, and training of students. The workshop attendees strongly encourage NSF to continue and extend the program.

Building on that success, the workshop highlighted a few important classes of Internet measurement research for which the initial IMR program was not a good fit. The IMR program funded prototyping/development of tools and methodologies, and a theme of the discussion at the workshop was the extension of support to other stages of measurement, such as deployment, maintenance of tools and datasets, scaling, and community development, as well as iteration over time on all phases to make sure that the tools/methods keep pace with Internet changes and with stakeholder needs.

Longitudinal measurements: The IMR program supported three award tracks: Track 1: Methodologies and Methods (MM), Track 2: Measurement Tool Development and Demonstration (MT), and Track 3: Internet Measurement Related Infrastructure-Planning (RI-P). The tracks devoted to deploying measurement infrastructure (RI-P) and tools (MT) had project lengths of 1 and 2 years, respectively, too short of a time to support longitudinal studies, which, as highlighted above, are extremely important to understanding and improving the always-evolving Internet. The MM track had project lengths of 3-4 years, but it focused less on measurement collection than on methodology devel-

opment, and even 3-4 years is a relatively short time period for longitudinal study, especially when factoring in the time in the project that will need to be devoted to developing the new methodologies before they are ready to support quality measurements. Going forward, there would be great benefit to funding opportunities that fund an initial development/research period at funding levels and durations similar to IMR and other existing programs, then fund an extended maintenance and data collection period at a lower level, perhaps as a recurring supplement opportunity funding 1-2 years at a time.

Larger projects: The IMR MM and MT tracks allowed awards of up to \$600,000, an amount that most often enables single institutions and often single PI budgets. Given the multi-dimensional, multi-stakeholder nature of Internet measurement, as well as the need for an array of vantage points and measurement techniques to capture aspects of the Internet with wide coverage, there is benefit to awards with larger budgets to encourage and fund more widespread collaboration. Larger awards would further enable both building tools and measurement systems, validating their accuracy, and then analyzing and disseminating results beyond the academic research community.

Infrastructure: Given the need for distributed vantage points and longitudinal measurement and data-sharing, there is value in tracks to support such efforts. The infrastructure that supports Internet measurement research is often complex and requires ongoing maintenance and support to guarantee quality data. Support for maintenance will multiply the impact of the program.